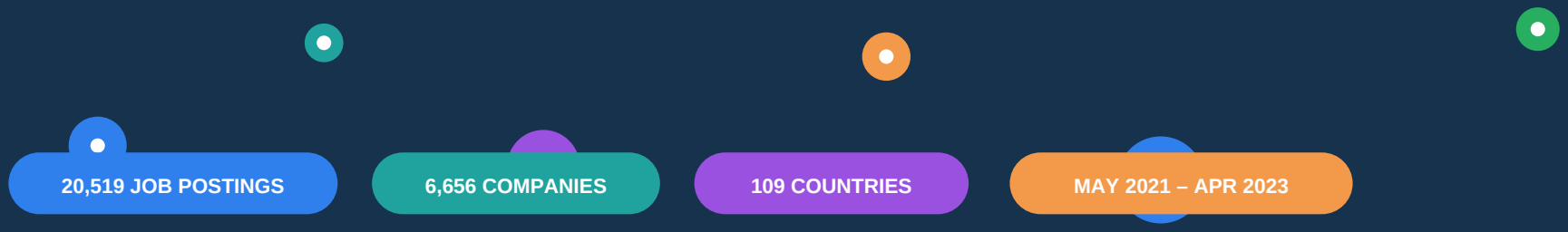
LINKEDIN JOB POSTINGS

SQL Analytics Project

A pictorial walkthrough of the data-cleaning pipeline, SQL Server star schema, analysis techniques, report views, and business insights.



Visual project guide

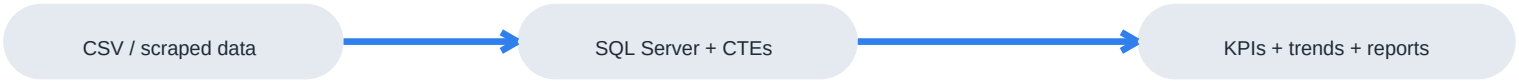
1. What is this project?

Turning raw LinkedIn job postings into decision-ready hiring insights

20,519	6,656	109	11.3%
job postings	companies	countries	data/ML/analytics roles

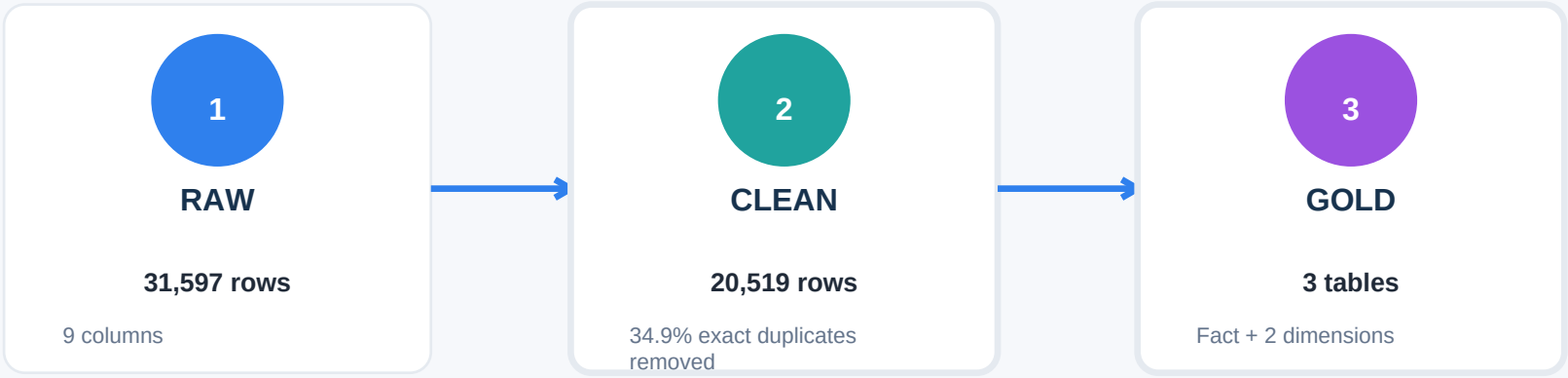
BUSINESS QUESTION

What does the LinkedIn hiring market look like across companies, industries, countries, seniority levels and time — and where are the strongest opportunities?



2. Data cleaning: from messy to usable

Python/pandas prepares the dataset before SQL analytics

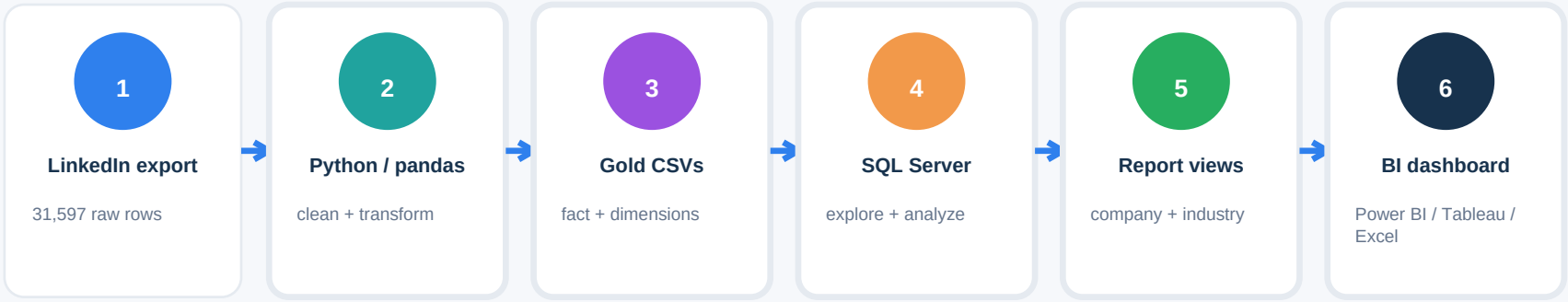


CLEANING ACTIONS

- **Whitespace / HTML** → strip unwanted text artifacts
- **Duplicates** → remove exact duplicate rows
- **Hiring status** → split status from benefit count
- **Seniority** → null out mis-scraped values
- **Location** → parse city / region / country

3. End-to-end architecture

The complete pipeline from scraped data to dashboards



WHY THIS ARCHITECTURE?

Separating cleaning, storage, analysis and reporting makes the project easier to test, reuse and extend. SQL analysts work on a structured gold layer instead of repeatedly cleaning raw data.

4. SQL Server data model: star schema

One central fact table connected to company and location dimensions



Why a star schema?

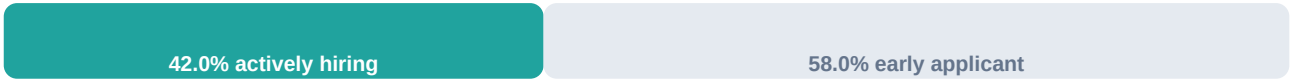
Faster analysis • cleaner joins • reusable dimensions

5. Headline KPIs

The first SQL questions turn 20K+ rows into a market snapshot

20,519	8,616	42%	2,311
Total postings	Actively hiring	Active-hiring rate	Data / ML / Analytics

Hiring status mix

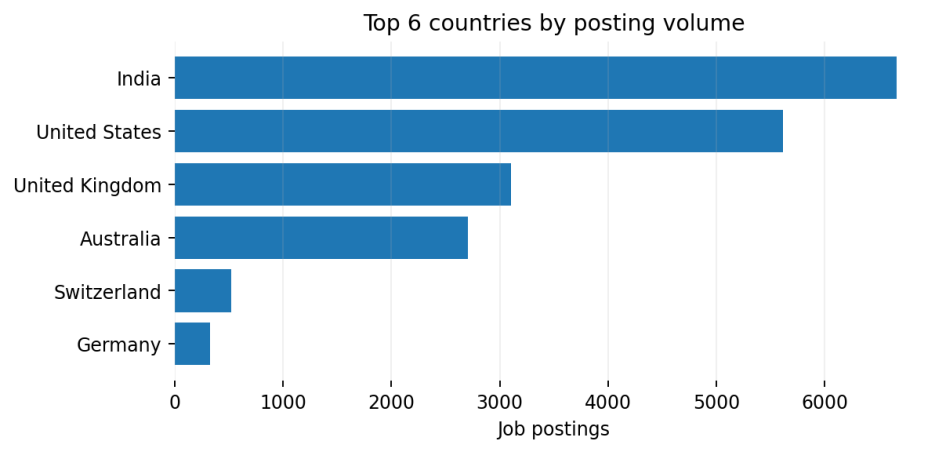


Average benefits listed per posting: 0.11

Date range: May 2021 → Apr 2023

6. Geography: where are the jobs?

Country-level aggregation reveals strong market concentration



Key takeaway

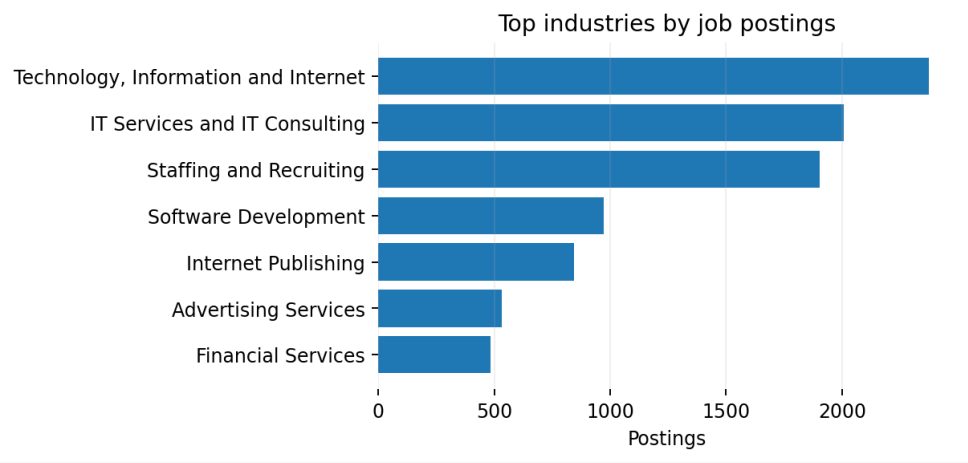
India leads the dataset, followed by the United States, United Kingdom and Australia.

SQL idea:
GROUP BY country

1. India	**6,666**
2. United States	**5,617**
3. United Kingdom	**3,102**
4. Australia	**2,704**

7. Industry ranking

TOP N + RANK() identifies the industries generating the most demand



Top 3 = ~31%

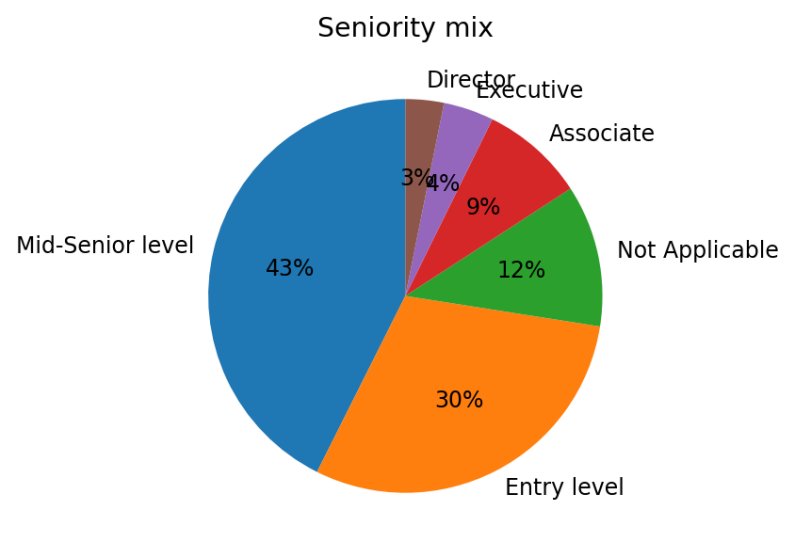
Technology, Information and Internet; IT Services and IT Consulting; Staffing and Recruiting dominate the dataset.

SQL techniques

- GROUP BY
- ORDER BY DESC
- TOP 5
- RANK() OVER()

8. Who are companies hiring?

Seniority distribution shows where the market is concentrated



Largest segment

Mid-Senior level

7,786 postings

≈ 42.6% of all postings

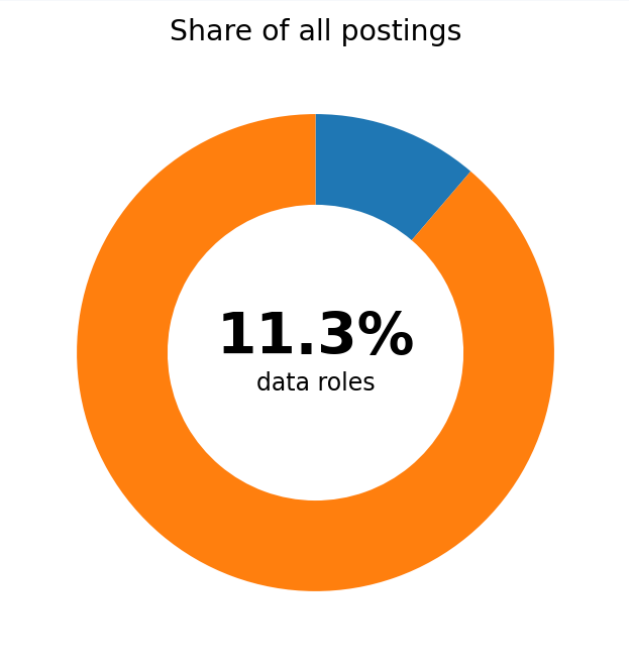
Next:

Entry level ≈ 26.6%

Useful for candidate targeting

9. Data / ML / Analytics opportunity

A focused flag turns a broad job dataset into a career-specific view



What the flag does

The is_data_role field allows SQL to isolate data, ML and analytics jobs from the wider market without creating a separate dataset.

2,311 identified roles

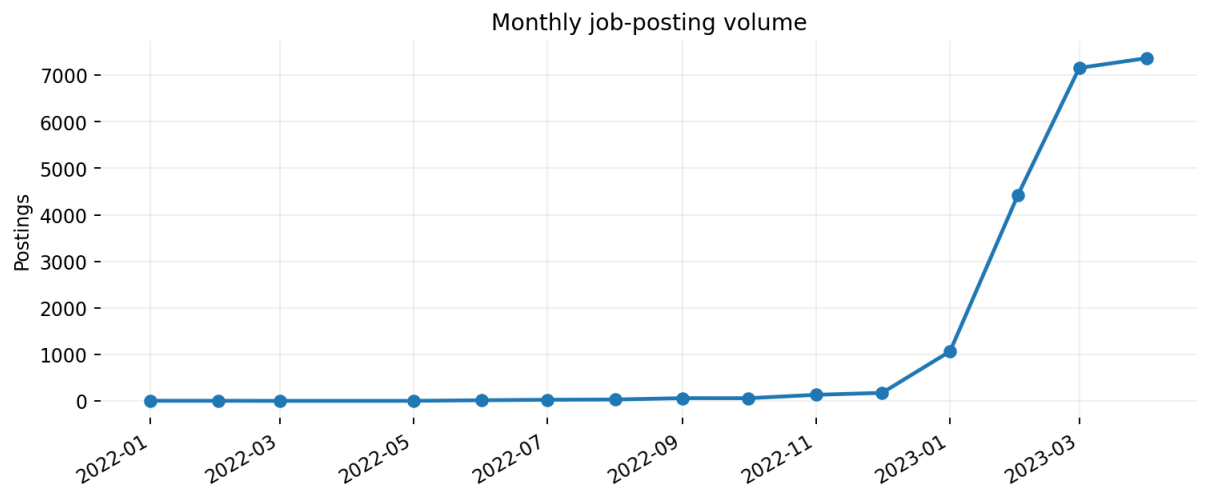
≈ 1 in every 9 postings

Great for:

- career targeting
- market sizing
- country comparison

10. Change over time

Monthly aggregation makes hiring activity visible as a time series



SQL

YEAR()

MONTH()

DATETRUNC()

FORMAT()

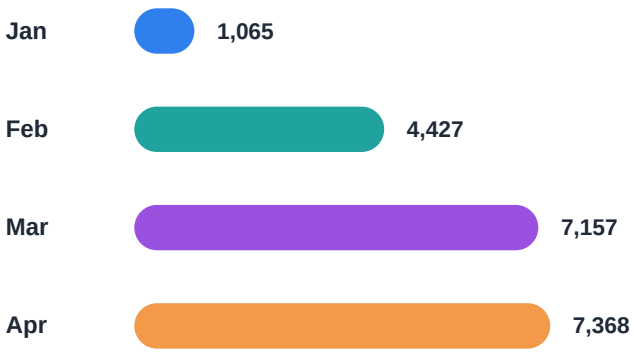
Use case

Spot growth, seasonality and unusual spikes.

11. Advanced SQL: cumulative & month-over-month analysis

Window functions turn simple counts into performance signals

LAG() = compare with previous month



current - previous = MoM change

Window functions

1. **Monthly count**
   COUNT(*)
2. **Previous month**
   LAG(count)
3. **Running total**
   SUM() OVER
4. **Moving average**
   AVG() OVER

12. From analysis to reusable report views

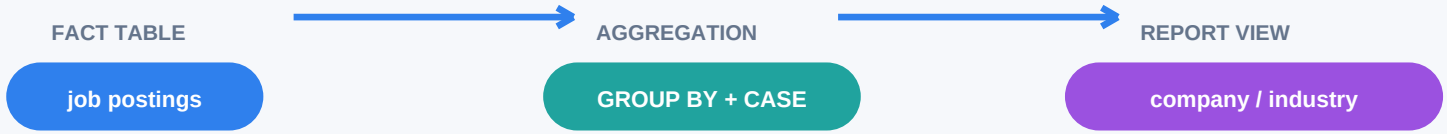
The project converts raw aggregates into business-friendly company and industry segments

Company segmentation

- 20+ — High-Volume Recruiter
- 3–19 — Regular Employer
- 1–2 — One-Off Poster

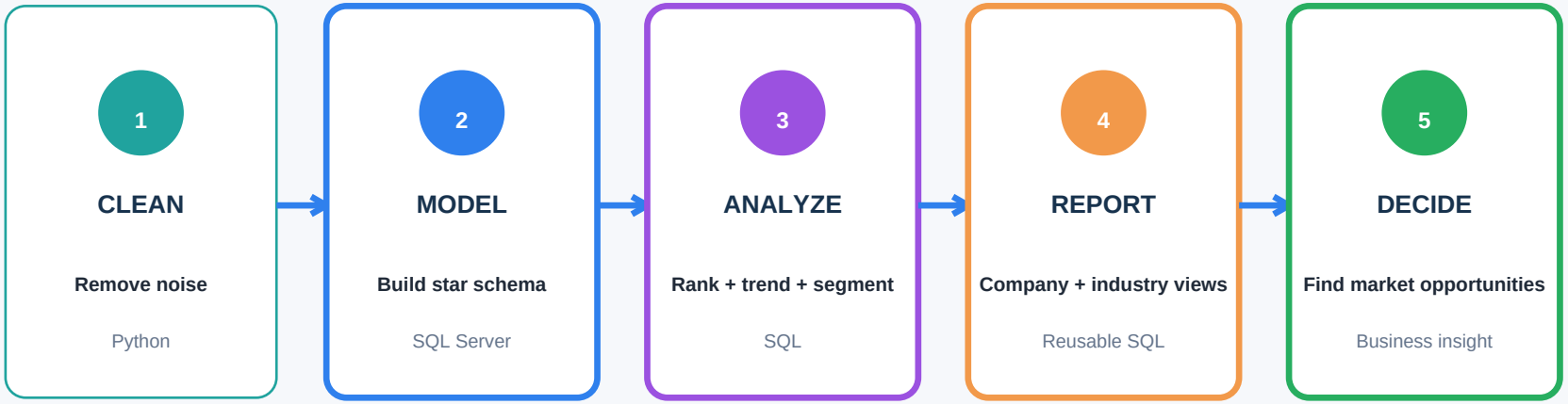
Industry segmentation

- >1000 — High-Demand
- 200–1000 — Mid-Demand
- <200 — Low-Demand



13. The project story in one picture

A recruiter can understand the value chain in under a minute



Key business takeaways

- India is the largest market in this dataset (6,666 postings).
- Mid-Senior roles are the largest seniority group (~37.9%).
- Data / ML / Analytics roles represent ~11.3% of postings.
- Only ~42% are marked Actively Hiring — useful context when interpreting raw job volume.

SQL skills demonstrated: CTEs • JOINs • GROUP BY • CASE • TOP • RANK() • LAG() • window functions • date functions • reusable views

14. SQL technique cheat sheet

What each script contributes to the project

01	**Database exploration**	INFORMATION_SCHEMA
02	**Dimensions exploration**	DISTINCT / profiling
03	**Date range**	MIN / MAX
04	**Measures**	COUNT / SUM / AVG
05	**Magnitude**	GROUP BY
06	**Ranking**	TOP / RANK
07	**Time series**	DATETRUNC / FORMAT
08	**Cumulative**	SUM() OVER / moving average
09	**Performance**	LAG() / MoM
10	**Segmentation**	CASE
11	**Part-to-whole**	% of total
12–13	**Reports**	CREATE VIEW

Result: a portfolio project that demonstrates an end-to-end analytics workflow, not just isolated SQL queries.